MA 465: Introduction to Mathematical Modeling

Final Project

Models of the Anderson Iris Data

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1. Abstract

This report examines the Iris dataset with the goal of modeling sepal length as a function of sepal width, petal length, and petal width across the three species: Setosa, Versicolor, and Virginica. A multiple linear regression model was used to estimate the relationships between these variables, along with the associated parameter uncertainties. The significance of each predictor was evaluated using t-tests, and Pearson correlation coefficients were computed to assess the strength of the relationships between sepal length and petal length.

The results indicate that petal width is the most consistently significant predictor for Versicolor and Virginica, while sepal width has a stronger influence for Setosa. Petal length was generally not statistically significant in the multivariate model, likely due to correlations with other predictors. Correlation analysis shows weak linear relationships for Setosa and stronger, statistically significant relationships for Versicolor and Virginica. A chi-square goodness-of-fit test further indicates that the model's performance is sensitive to the assumed noise variance, with acceptable fits occurring only under more realistic noise assumptions.

Overall, linear models provide a useful first approximation of the relationships within the dataset, but their effectiveness varies across species. These results suggest that species-specific models or more advanced nonlinear approaches may better represent the underlying structure of the data.

2. Introduction

The purpose of this report is to analyze the Anderson Iris dataset using a variety of statistical modeling techniques, with a focus on understanding the relationships between sepal length and other measured features. Specifically, multi-variable linear regression, goodness-of-fit analysis, and simple linear regression with correlation assessment are applied separately to each of the three Iris species. These methods provide a framework for evaluating both the predictive capability of linear models and the statistical significance of individual variables.

First, multi-variable linear regression is used to model sepal length as a function of sepal width, petal length, and petal width. The resulting model parameters and their associated uncertainties are computed for each species, providing insight into how each independent variable contributes to the overall model. Next, the quality of these models is assessed through statistical measures including t-tests, chi-square analysis, and the coefficient of determination (R^2), allowing for evaluation of both parameter significance and overall fit quality under different assumptions of measurement uncertainty.

In addition, a simpler ordinary linear regression model is explored by examining the relationship between sepal length and petal length. This analysis includes estimation of regression coefficients, their uncertainties, and the calculation of the linear correlation coefficient to determine the strength and significance of the relationship.

Finally, the report synthesizes these results to evaluate the effectiveness of linear modeling for this dataset. Based on quantitative evidence, conclusions are drawn regarding the suitability of the chosen models and whether alternative modeling approaches may be warranted. Overall, this study aims to provide a comprehensive statistical characterization of the Iris dataset and to highlight key considerations in model selection and evaluation.

3. Multi-Variable Linear Regression

3.1 Methodology

A multi-variable linear regression model was constructed for each of the three Iris species (setosa, versicolor, and virginica) using the sepal length (SL) as the dependent variable. The three remaining measurements—sepal width (SW), petal length (PL), and petal width (PW)—were used as independent regressors. The model assumes a linear relationship of the form $y = a_0 + a_1x_1 + a_2x_2 + a_3x_3$ where the coefficients a_k were determined using the least-squares solution implemented in MATLAB via the backslash operator.

To estimate the uncertainty in these fitted parameters, the spread of the data around the regression model was analyzed using the covariance of the coefficients. Specifically, the parameter errors were obtained from the matrix expression $cov(a) = \sigma^2(X^TX)^{-1}$ where σ^2 is estimated from the residuals of the fit. The standard error of each coefficient is then given by the square root of the diagonal entries of this covariance matrix. These values provide a measure of how sensitive each fitted parameter is to variations in the data and were recorded alongside the coefficients for subsequent statistical analysis.

3.2 Results

Table 1 shows the best-fit parameters and their errors for each of the three Iris species. The results were calculated using the MATLAB code found in Appendix A.

Species	Parameter	Best-Fit Value	Error
Setosa	a_0 (SW)	2.3519	0.3929
	a_1 (SL)	0.6548	0.0924
	a_2 (PW)	0.2376	0.2080
	a_3 (PL	0.2521	0.3469
Versicolor	a_0 (SW)	1.8955	0.5071
	a_1 (SL)	0.3969	0.2045
	a_2 (PW)	0.9083	0.1654
	a_3 (PL	-0.6792	0.4354
Virginica	a_0 (SW)	0.6999	0.5336
	a_1 (SL)	0.3303	0.1743
	a_2 (PW)	0.9455	0.0907
	a_3 (PL	-0.1698	0.1981

Table 1: Multiple Linear Regression Best-Fit Values and Errors

For Setosa, the largest contribution to sepal length comes from sepal width (), while petal length and petal width have relatively smaller coefficients. However, the large uncertainty associated with the petal length coefficient (±0.3469) suggests that this variable is not a reliable predictor for this species. This indicates that Setosa's sepal length is primarily influenced by sepal width rather than petal-based features.

In contrast, Versicolor and Virginica show a different trend. In both species, petal width () has the strongest positive influence on sepal length, with relatively small uncertainties, indicating a stable and significant relationship. Petal length, however, shows a weaker and even negative contribution in both cases, suggesting that when other variables are accounted for, it does not independently increase sepal length and may instead be correlated with other predictors.

The intercept values decrease progressively from Setosa to Virginica, reflecting baseline differences in sepal length between species. Additionally, the generally smaller uncertainties in the petal width coefficients for Versicolor and Virginica suggest that this feature is the most consistently informative predictor in the multivariate model.

Overall, these results indicate that while all three species exhibit relationships between physical features, the dominant predictors differ by species, with Setosa being more influenced by sepal width and the other two species being more strongly governed by petal width.

4. Goodness of Fit

4.1 Methodology

To evaluate the importance of each regressor in the linear models, t-statistics were computed for every fitted parameter using the ratio $t_k = \frac{a_k}{SE(a_k)}$ where a_k is the estimated coefficient and $SE(a_k)$ is its corresponding standard error obtained from Part 1. These t-values quantify how large each coefficient is relative to its uncertainty. A two-tailed hypothesis test was then performed at the 99% confidence level by comparing the absolute value of each t-statistic to a critical value. The t_{crit} values were obtained by using MATLAB's tinv function with the confidence level (0.995 for two-tailed test) and the degrees of freedom as input values. Parameters with $|t_k| > t_{crit}$ were considered statistically significant and therefore necessary in the model, while those below this threshold were deemed not significant and potentially removable. This procedure was applied independently to each species to identify the most relevant predictors of sepal length.

To assess the goodness of fit of the regression models, a chi-square (χ^2) test was performed for each iris species under the assumption of a uniform measurement uncertainty. The chi-square statistic was computed as $\chi_{\min} = \sum \frac{(y - X\mathbf{a})^2}{\sigma^2}$, where y is the observed sepal length, X is the design matrix, $\mathbf{a}$ is the vector of fitted regression coefficients, and σrepresents the assumed standard deviation of measurement noise. This procedure was carried out twice, using $\sigma = 0.1$mm and $\sigma = 0.3$mm, to evaluate the sensitivity of the model to measurement uncertainty. For each case, the computed $\chi_{\min}$was compared against the corresponding critical chi-square value at the 99% confidence level with the appropriate degrees of freedom. A model was considered statistically acceptable if $\chi_{\min} < \chi^2_{\text{crit}}$, indicating that the observed deviations could be explained by the assumed noise level. This approach was applied independently to each species to determine whether the linear regression model provided a statistically consistent description of the data.

4.2 Results

The t_{crit} value for all three Iris species was calculated to be $t_{crit} = 2.687$. The t-test results and summary of significant values can be found in Table 2 below.

Species	Parameter	T-Test Value	Significant?
Setosa	a_0 (SW)	5.986	---
	a_1 (SL)	7.083	Yes
	a_2 (PW)	1.142	No
	a_3 (PL)	0.727	No
Versicolor	a_0 (SW)	3.738	---
	a_1 (SL)	1.891	No
	a_2 (PW)	5.491	Yes
	a_3 (PL)	-1.560	No
Virginica	a_0 (SW)	1.312	---
	a_1 (SL)	1.895	No
	a_2 (PW)	10.422	Yes
	a_3 (PL)	-0.857	No

Table 2: T-Test and Significance

Tables 3 and 4 show the chi-squared goodness of fit using $\sigma = 0.1$ in Table 3 and $\sigma = 0.3$ in Table 4.

Species	χ^2_{min}	χ^2_{crit} (99%)	Acceptable Fit?
Setosa	258.66	71.20	No
Veriscolor	515.64	71.20	No
Virginica	465.17	71.20	No

Table 3: Chi-Squared Goodness of Fit ($\sigma = 0.1$)

Species	χ^2_{min}	χ^2_{crit} (99%)	Acceptable Fit?
Setosa	28.74	71.20	Yes
Veriscolor	57.29	71.20	Yes
Virginica	51.69	71.20	Yes

Table 3: Chi-Squared Goodness of Fit ($\sigma = 0.3$)

For Setosa, only the sepal length coefficient (a_1) is statistically significant, with a t-value of 7.083, well above the critical threshold. In contrast, sepal width and petal-based features are not significant, suggesting that the model primarily captures variation in sepal length through sepal length–related structure rather than the full set of predictors.

For Versicolor, petal width (a_2) is the only statistically significant predictor, with a t-value of 5.491. This indicates that petal width plays a dominant role in explaining sepal length for this species, while the remaining variables do not contribute significantly when considered jointly in the regression model.

Similarly, for Virginica, petal width again emerges as the only significant predictor, with a very large t-value of 10.422. This strongly suggests that petal width is the most robust explanatory variable for sepal length in this species across the dataset.

Across all three species, petal length does not appear as a statistically significant predictor once other variables are included in the model. This suggests that its effect may be redundant or absorbed by correlated variables such as petal width.

5. Correlation Coefficient

5.1 Methodology

For each Iris species (Setosa, Versicolor, and Virginica), ordinary least squares (OLS) linear regression was performed to model the relationship between sepal length y and petal length x. The model assumed a linear form $y = a + b$ where a is the intercept and b is the slope. The regression coefficients were computed using the matrix least-squares solution:

$$\boldsymbol{a} = (X^T X)^{-1} X^T y$$

where X=[1,x] is the design matrix containing a column of ones for the intercept and the predictor variable x.

Residuals were computed as $r = y - X\boldsymbol{a}$ and the noise variance was estimated using:

$$\sigma^2 = \frac{1}{N-p} \sum r^2$$

where N is the number of observations and p=2 is the number of parameters. The covariance matrix of the estimated coefficients was then calculated as:

$$Cov(\boldsymbol{a}) = \sigma^2 (X^T X)^{-1}$$

from which standard errors were obtained as the square roots of the diagonal elements.

To quantify linear dependence between variables, the Pearson correlation coefficient was computed:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

The statistical significance of the correlation was tested using a t-test under the null hypothesis H with the t-test statistic:

$$t = r \sqrt{\frac{N-2}{1-r^2}}$$

This value was compared against the critical t-value $t_{crit} = t_{0.975,N-2}$ corresponding to a 95% confidence level (two-tailed test). All computations were implemented in MATLAB using built-in linear algebra operations and statistical functions such as \, corr, and tinv.

5.2 Results

Tables 4 and 5 show the estimated parameters from the linear regression models, their associated uncertainties, and the statistical significance of the correlation analysis for each Iris species. Figure 1 displays scatter plots of sepal length versus petal length for for each of the three Iris species along with the corresponding best-fit regression lines obtained using ordinary least squares.

Species	Parameter	Best-Fit Value	Error
Setosa	a	4.2132	0.4156
	b	0.5423	0.2823
Versicolor	a	2.4075	0.4463
	b	0.8283	0.1041
Virginica	a	1.0597	0.4668
	b	0.9957	0.0837

Table 4: Linear Regression Values and Errors

Species	Correlation r	T-Value	t_{crit}	Significant?
Setosa	0.2672	1.9209	2.0106	No
Versicolor	0.7540	7.9538		Yes
Virginica	0.8642	11.9011		Yes

Table 4: Correlation and Significance Test

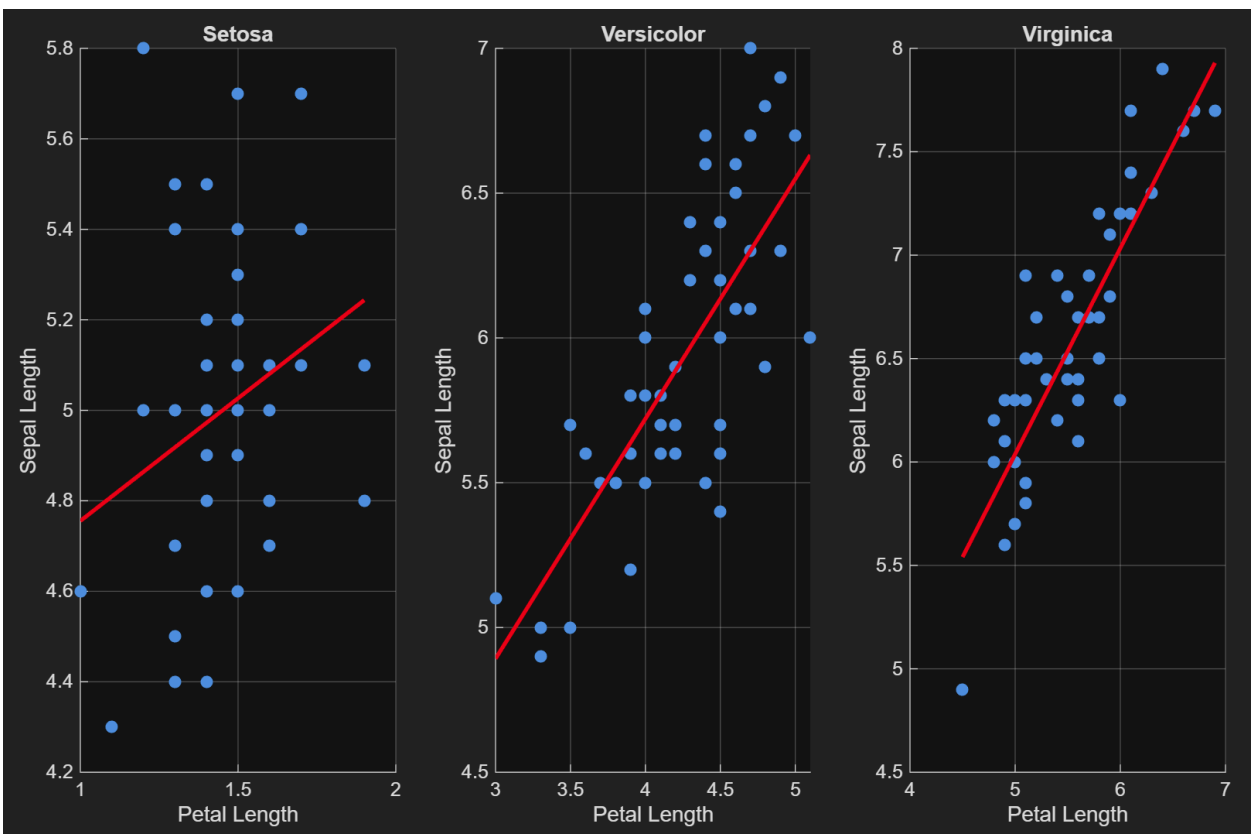

Figure 1: Sepal Length vs Petal Length

From the regression results, all three species exhibit a positive relationship between petal length and sepal length, as indicated by the positive slope values in each case. However, the strength of this relationship varies significantly across species. Setosa shows a relatively weak slope and large uncertainty, indicating that petal length is not a strong predictor of sepal length for this species. This is supported by the low correlation coefficient (r=0.2672) and the fact that the relationship is not statistically significant at the 95% confidence level.

In contrast, both Versicolor and Virginica display much stronger linear behavior. Versicolor shows a moderate to strong correlation (r=0.7540), while Virginica exhibits a very strong correlation (r=0.8642). In both cases, the computed t-statistics exceed the critical value, confirming that the correlations are statistically significant at the 95% confidence level. This indicates that petal length is a reliable predictor of sepal length for these species.

Overall, the results demonstrate a clear species-dependent difference in the relationship between physical features. Virginica shows the strongest linear dependence, followed by Versicolor, while Setosa shows the weakest. These trends are also reflected visually in the regression plots, where the data points for Virginica and Versicolor align closely with the fitted line, whereas Setosa shows greater scatter around the model.

6. Modeling Lessons

Based on the results from both the multiple linear regression and the correlation analysis, the most appropriate linear model for predicting sepal length is a species-dependent multivariate model of the form:

$$y = a_0 + a_1(\text{Sepal Width}) + a_2(\text{Petal Width}) + a_3(\text{Petal Length})$$

However, the statistical significance results (Table 2) indicate that not all predictors contribute meaningfully for each species. In particular, petal width consistently emerges as the most significant and reliable predictor for Versicolor and Virginica, while Setosa shows stronger dependence on sepal width. Petal length is generally not statistically significant once the other variables are included, suggesting redundancy due to correlation between predictors.

Therefore, the “best” linear model in a practical sense would be a reduced model, where only statistically significant predictors are retained for each species. This leads to a simplified structure such as:

- Setosa: primarily dependent on sepal width
- Versicolor: primarily dependent on petal width
- Virginica: primarily dependent on petal width

This reduced model improves interpretability while maintaining statistical validity, as it removes variables that do not pass the t-test significance threshold.

The results indicate that a linear model is partially appropriate but not universally optimal for describing the full dataset.

From the correlation analysis, strong linear relationships are observed for Versicolor and Virginica ($r=0.7540$ and $r=0.8642$, respectively), both of which are statistically significant at the 95% confidence level. This supports the assumption that a linear model captures a meaningful trend for these species. However, Setosa exhibits a weak correlation ($r=0.2672$) that is not statistically significant, indicating that a linear relationship is not strongly supported in this case.

Additionally, the chi-square goodness-of-fit results show strong dependence on the assumed noise level. When $\sigma=0.1$, all models are rejected due to excessively large residual errors, suggesting that the linear model underestimates the variability in the data. When $\sigma=0.3$, all models become acceptable, indicating that the linear model can be reasonable under a more realistic noise assumption.

However, the presence of non-significant coefficients, multicollinearity between predictors, and species-dependent behavior suggests that a single global linear model is not fully sufficient. These limitations imply that while linear regression is useful for capturing first-order trends, it may not fully describe the complexity of the relationships in the dataset.

Therefore, it would be reasonable to explore alternative models, such as non-linear regression models (e.g., polynomial terms) or species-specific modeling approaches.

In summary, the linear model is useful as a first-order approximation, but not necessarily the most complete or robust representation of the underlying data structure.

7. Conclusion

This project applied multiple statistical techniques to model sepal length in the Iris dataset as a function of sepal width, petal length, and petal width across three species. Using multiple linear regression, parameter estimates and uncertainties were obtained, and the significance of each predictor was evaluated using t-tests. The results showed that petal width is the most consistently significant predictor for Versicolor and Virginica, while Setosa is more weakly modeled by the chosen predictors, with fewer statistically meaningful contributions.

Correlation analysis between sepal length and petal length further highlighted species-dependent differences in linear structure. Versicolor and Virginica exhibited strong,

statistically significant correlations, whereas Setosa showed a weak and non-significant relationship. These findings were supported by chi-square goodness-of-fit testing, which demonstrated that model acceptability depends strongly on the assumed noise level, with reasonable fits observed only under more realistic variance assumptions.

Overall, the results indicate that linear regression provides a useful first-order approximation of the relationships in the dataset, particularly for Versicolor and Virginica. However, the variability in statistical significance across predictors and species suggests that a single unified linear model is limited in its descriptive power. Future work could improve model performance by incorporating nonlinear terms, performing feature selection, or developing species-specific models to better capture the underlying structure of the data.

8. Appendices

Appendix A: Complete MATLAB code for graphing and calculations:

```
% Iris Data [SL SW PL PW]
Iris_setosa = [
5.1 3.5 1.4 0.2; 4.9 3.0 1.4 0.2; 4.7 3.2 1.3 0.2; 4.6 3.1 1.5 0.2; 5.0 3.6 1.4 0.2;
5.4 3.9 1.7 0.4; 4.6 3.4 1.4 0.3; 5.0 3.4 1.5 0.2; 4.4 2.9 1.4 0.2; 4.9 3.1 1.5 0.1;
5.4 3.7 1.5 0.2; 4.8 3.4 1.6 0.2; 4.8 3.0 1.4 0.1; 4.3 3.0 1.1 0.1; 5.8 4.0 1.2 0.2;
5.7 4.4 1.5 0.4; 5.4 3.9 1.3 0.4; 5.1 3.5 1.4 0.3; 5.7 3.8 1.7 0.3; 5.1 3.8 1.5 0.3;
5.4 3.4 1.7 0.2; 5.1 3.7 1.5 0.4; 4.6 3.6 1.0 0.2; 5.1 3.3 1.7 0.5; 4.8 3.4 1.9 0.2;
5.0 3.0 1.6 0.2; 5.0 3.4 1.6 0.4; 5.2 3.5 1.5 0.2; 5.2 3.4 1.4 0.2; 4.7 3.2 1.6 0.2;
4.8 3.1 1.6 0.2; 5.4 3.4 1.5 0.4; 5.2 4.1 1.5 0.1; 5.5 4.2 1.4 0.2; 4.9 3.1 1.5 0.2;
5.0 3.2 1.2 0.2; 5.5 3.5 1.3 0.2; 4.9 3.6 1.4 0.1; 4.4 3.0 1.3 0.2; 5.1 3.4 1.5 0.2;
5.0 3.5 1.3 0.3; 4.5 2.3 1.3 0.3; 4.4 3.2 1.3 0.2; 5.0 3.5 1.6 0.6; 5.1 3.8 1.9 0.4;
4.8 3.0 1.4 0.3; 5.1 3.8 1.6 0.2; 4.6 3.2 1.4 0.2; 5.3 3.7 1.5 0.2; 5.0 3.3 1.4 0.2
];

Iris_versicolor = [
7.0 3.2 4.7 1.4; 6.4 3.2 4.5 1.5; 6.9 3.1 4.9 1.5; 5.5 2.3 4.0 1.3; 6.5 2.8 4.6 1.5;
5.7 2.8 4.5 1.3; 6.3 3.3 4.7 1.6; 4.9 2.4 3.3 1.0; 6.6 2.9 4.6 1.3; 5.2 2.7 3.9 1.4;
5.0 2.0 3.5 1.0; 5.9 3.0 4.2 1.5; 6.0 2.2 4.0 1.0; 6.1 2.9 4.7 1.4; 5.6 2.9 3.6 1.3;
6.7 3.1 4.4 1.4; 5.6 3.0 4.5 1.5; 5.8 2.7 4.1 1.0; 6.2 2.2 4.5 1.5; 5.6 2.5 3.9 1.1;
5.9 3.2 4.8 1.8; 6.1 2.8 4.0 1.3; 6.3 2.5 4.9 1.5; 6.1 2.8 4.7 1.2; 6.4 2.9 4.3 1.3;
6.6 3.0 4.4 1.4; 6.8 2.8 4.8 1.4; 6.7 3.0 5.0 1.7; 6.0 2.9 4.5 1.5; 5.7 2.6 3.5 1.0;
5.5 2.4 3.8 1.1; 5.5 2.4 3.7 1.0; 5.8 2.7 3.9 1.2; 6.0 2.7 5.1 1.6; 5.4 3.0 4.5 1.5;
6.0 3.4 4.5 1.6; 6.7 3.1 4.7 1.5; 6.3 2.3 4.4 1.3; 5.6 3.0 4.1 1.3; 5.5 2.5 4.0 1.3;
5.5 2.6 4.4 1.2; 6.1 3.0 4.6 1.4; 5.8 2.6 4.0 1.2; 5.0 2.3 3.3 1.0; 5.6 2.7 4.2 1.3;
5.7 3.0 4.2 1.2; 5.7 2.9 4.2 1.3; 6.2 2.9 4.3 1.3; 5.1 2.5 3.0 1.1; 5.7 2.8 4.1 1.3
];
```

```
Iris_virginica = [
6.3 3.3 6.0 2.5; 5.8 2.7 5.1 1.9; 7.1 3.0 5.9 2.1; 6.3 2.9 5.6 1.8; 6.5 3.0 5.8 2.2;
7.6 3.0 6.6 2.1; 4.9 2.5 4.5 1.7; 7.3 2.9 6.3 1.8; 6.7 2.5 5.8 1.8; 7.2 3.6 6.1 2.5;
6.5 3.2 5.1 2.0; 6.4 2.7 5.3 1.9; 6.8 3.0 5.5 2.1; 5.7 2.5 5.0 2.0; 5.8 2.8 5.1 2.4;
6.4 3.2 5.3 2.3; 6.5 3.0 5.5 1.8; 7.7 3.8 6.7 2.2; 7.7 2.6 6.9 2.3; 6.0 2.2 5.0 1.5;
6.9 3.2 5.7 2.3; 5.6 2.8 4.9 2.0; 7.7 2.8 6.7 2.0; 6.3 2.7 4.9 1.8; 6.7 3.3 5.7 2.1;
7.2 3.2 6.0 1.8; 6.2 2.8 4.8 1.8; 6.1 3.0 4.9 1.8; 6.4 2.8 5.6 2.1; 7.2 3.0 5.8 1.6;
7.4 2.8 6.1 1.9; 7.9 3.8 6.4 2.0; 6.4 2.8 5.6 2.2; 6.3 2.8 5.1 1.5; 6.1 2.6 5.6 1.4;
7.7 3.0 6.1 2.3; 6.3 3.4 5.6 2.4; 6.4 3.1 5.5 1.8; 6.0 3.0 4.8 1.8; 6.9 3.1 5.4 2.1;
6.7 3.1 5.6 2.4; 6.9 3.1 5.1 2.3; 5.8 2.7 5.1 1.9; 6.8 3.2 5.9 2.3; 6.7 3.3 5.7 2.5;
6.7 3.0 5.2 2.3; 6.3 2.5 5.0 1.9; 6.5 3.0 5.2 2.0; 6.2 3.4 5.4 2.3; 5.9 3.0 5.1 1.8
];

% Part 1 ---------------------------------------------------------------------------
% Section A + B (coefficients + errors)

% Setosa
y = Iris_setosa(:,1);
X = [ones(size(Iris_setosa,1),1) Iris_setosa(:,2:4)];
[a_setosa, se_setosa] = linreg_with_errors(X, y)

% Versicolor
y = Iris_versicolor(:,1);
X = [ones(size(Iris_versicolor,1),1) Iris_versicolor(:,2:4)];
[a_versicolor, se_versicolor] = linreg_with_errors(X, y)

% Virginica
y = Iris_virginica(:,1);
X = [ones(size(Iris_virginica,1),1) Iris_virginica(:,2:4)];
[a_virginica, se_virginica] = linreg_with_errors(X, y)

% Part 2 ---------------------------------------------------------------------------
% (a) t-statistics + 99% significance test

% Degrees of freedom
df_setosa = size(Iris_setosa,1) - 4;
df_versicolor = size(Iris_versicolor,1) - 4;
df_virginica = size(Iris_virginica,1) - 4;

% t-statistics
t_setosa = a_setosa ./ se_setosa;
t_versicolor = a_versicolor ./ se_versicolor;
t_virginica = a_virginica ./ se_virginica;
```

```
% Critical t-values (two-tailed, 99% confidence)
tcrit_setosa = tinv(0.995, df_setosa)
tcrit_versicolor = tinv(0.995, df_versicolor)
tcrit_virginica = tinv(0.995, df_virginica)

% Significance masks (true = needed regressor)
sig_setosa = abs(t_setosa) > tcrit_setosa;
sig_versicolor = abs(t_versicolor) > tcrit_versicolor;
sig_virginica = abs(t_virginica) > tcrit_virginica;

% Display results in readable format
labels = ["Intercept", "Sepal Width", "Petal Length", "Petal Width"];

disp('setosa');
for i = 1:4
   fprintf('%s: t = %.3f, significant = %d\n', labels(i), t_setosa(i), sig_setosa(i));
end

disp('versicolor');
for i = 1:4
   fprintf('%s: t = %.3f, significant = %d\n', labels(i), t_versicolor(i), sig_versicolor(i));
end

disp('virginica');
for i = 1:4
   fprintf('%s: t = %.3f, significant = %d\n', labels(i), t_virginica(i), sig_virginica(i));
end

% (b) Chi-square goodness of fit test

sigmas = [0.1, 0.3];

% Function to compute chi-square
chi2_func = @(y, X, a, sigma) sum(((y - X*a).^2) / sigma^2);

% Degrees of freedom (same as before)
df_setosa = size(Iris_setosa,1) - 4;
df_versicolor = size(Iris_versicolor,1) - 4;
df_virginica = size(Iris_virginica,1) - 4;

% Loop over sigma values
for s = 1:length(sigmas)
   sigma = sigmas(s);
```

```
    % --- Setosa ---
    y = Iris_setosa(:,1);
    X = [ones(size(Iris_setosa,1),1) Iris_setosa(:,2:4)];
    chi2_setosa = chi2_func(y, X, a_setosa, sigma);

    % --- Versicolor ---
    y = Iris_versicolor(:,1);
    X = [ones(size(Iris_versicolor,1),1) Iris_versicolor(:,2:4)];
    chi2_versicolor = chi2_func(y, X, a_versicolor, sigma);

    % --- Virginica ---
    y = Iris_virginica(:,1);
    X = [ones(size(Iris_virginica,1),1) Iris_virginica(:,2:4)];
    chi2_virginica = chi2_func(y, X, a_virginica, sigma);

    % --- Critical chi-square (99%) ---
    chi2crit_setosa = chi2inv(0.99, df_setosa);
    chi2crit_versicolor = chi2inv(0.99, df_versicolor);
    chi2crit_virginica = chi2inv(0.99, df_virginica);

    % --- Accept/reject ---
    acc_setosa = chi2_setosa < chi2crit_setosa;
    acc_versicolor = chi2_versicolor < chi2crit_versicolor;
    acc_virginica = chi2_virginica < chi2crit_virginica;

    % --- Display ---
    fprintf('\n===== Sigma = %.1f =====\n', sigma);

    fprintf('Setosa:     chi2 = %.2f, crit = %.2f, acceptable = %d\n', ...
        chi2_setosa, chi2crit_setosa, acc_setosa);

    fprintf('Versicolor: chi2 = %.2f, crit = %.2f, acceptable = %d\n', ...
        chi2_versicolor, chi2crit_versicolor, acc_versicolor);

    fprintf('Virginica:  chi2 = %.2f, crit = %.2f, acceptable = %d\n', ...
        chi2_virginica, chi2crit_virginica, acc_virginica);
end

% Part 3 ------------------------------------------------------------------
% (a) Simple linear regression: y = a + b*x

figure;

species_names = {'Setosa','Versicolor','Virginica'};
```

```
datasets = {Iris_setosa, Iris_versicolor, Iris_virginica};

for k = 1:3

   data = datasets{k};

   x = data(:,3); % Petal Length
   y = data(:,1); % Sepal Length

   % Design matrix [1 x]
   X = [ones(length(x),1) x];

   % Coefficients
   a = X \ y;

   % Residuals
   r = y - X*a;

   N = length(y);
   p = 2;

   % Variance estimate
   sigma2 = sum(r.^2)/(N - p);

   % Covariance matrix
   CovA = sigma2 * inv(X' * X);

   % Standard errors
   se = sqrt(diag(CovA));

   fprintf('\n%s:\n', species_names{k});
   fprintf('Intercept (a) = %.4f ± %.4f\n', a(1), se(1));
   fprintf('Slope (b)    = %.4f ± %.4f\n', a(2), se(2));

   % Plot
   subplot(1,3,k);
   scatter(x,y,'filled'); hold on;

   x_fit = linspace(min(x), max(x), 100);
   y_fit = a(1) + a(2)*x_fit;

   plot(x_fit, y_fit, 'r', 'LineWidth', 2);

   title(species_names{k});
```

```
    xlabel('Petal Length');
    ylabel('Sepal Length');
    grid on;

    % --- Correlation coefficient ---
    r_corr = corr(x,y);

    % --- t-test for correlation ---
    t_stat = r_corr * sqrt((N - 2)/(1 - r_corr^2));
    t_crit = tinv(0.975, N - 2);

    significant = abs(t_stat) > t_crit;

    fprintf('Correlation r = %.4f\n', r_corr);
    fprintf('t = %.4f, t_crit = %.4f, significant (95%%) = %d\n', t_stat, t_crit, significant);
end

% Functions ------------------------------------------------------------------------------
function [a, se] = linreg_with_errors(X, y)

    % Fit coefficients
    a = X \ y;

    % Residuals
    r = y - X*a;

    N = length(y);
    p = size(X,2);

    % Estimate variance of noise
    sigma2 = sum(r.^2) / (N - p);

    % Covariance matrix of parameters
    CovA = sigma2 * inv(X' * X);

    % Standard errors
    se = sqrt(diag(CovA));

end
```